



Namal University, Mianwali

“University Timetable Management System”

Database Design Document

V1.0

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION:

The University Timetable Management System is a web-based application designed to centralize the creation and management of academic schedules within a university environment. In educational institutions, efficient time management is critical for both faculty and students. This system addresses the complexities of academic scheduling by providing a unified platform where administrators can create and publish timetables. Furthermore, it empowers faculty members to view their assigned schedules and formally submit schedule change requests, while allowing students to easily access their class timetables. By transitioning from manual, spreadsheet-based processes to a database-driven solution, the system aims to support the core administrative and academic activities of the university, ensuring a structured educational environment.

1.2. PROBLEM STATEMENT:

The current manual timetable scheduling process relies heavily on spreadsheets and informal communication channels such as emails, leading to significant inefficiencies. This approach frequently results in venue conflicts, where the same classroom is accidentally assigned to multiple courses simultaneously, and instructor conflicts, where a faculty member is scheduled to teach different classes at the same time. Additionally, the absence of a centralized data repository means that timetable information is scattered, making it difficult to maintain a single source of truth. These issues collectively lead to schedule mishaps and a poor experience for both staff and students.

1.3. PROJECT OBJECTIVES:

The primary objective of the University Timetable Management System is to develop a centralized, database-driven web application that streamlines academic scheduling.

Specific objectives include:

- Design and implement a normalized relational database schema to efficiently manage all entities related to university timetable.
- Enforce Role-Based Access Control (RBAC) to ensure secure and appropriate system access for Admin, Faculty, and Student users.
- Implement conflict detection mechanisms to prevent venue double-booking and instructor schedule overlaps.
- Enable administrators to create, edit, publish, and manage university timetables.
- Provide dedicated dashboards for faculty and students to view their respective assigned schedules and enrolled classes.
- Establish a formal workflow for faculty to submit schedule change requests.

1.4. GITHUB REPOSITORY LINK:

<https://github.com/hammad-013/university-timetable-management-system>

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY:

Sr. No	Entity Name	Description
01	User	An individual who has registered or created an account within the system.
02	Student	A specialized user representing a student enrolled in the university.
03	Faculty	A specialized user representing a teaching staff member in the university.
04	Department	An academic division within the university that offers various programs and courses.
05	Program	A specific degree offered by an academic department.
06	Batch	A specific group of students enrolled in a program during a particular academic year.
07	Semester	An academic term.
08	Session	An academic year that groups semester together.
09	Course	A subject offered by the university.
10	Venue	A physical location such as a classroom or laboratory, where classes are conducted.
11	Timeslot	A specific time period during a day of the week allocated for classes.
12	Schedule Change Request	A formal request submitted by a faculty member to modify an existing schedule assignment.
13	Section	A specific division of a batch used to organize students into smaller groups.

2.2. DATA DICTIONARY:

2.2.1. User:

Sr. No	Name	Data Type	Constraint	Description
01	user_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the user
02	email	String	NOT NULL, UNIQUE	Email address of the user
03	password_hash	String	NOT NULL	Encrypted password for authentication
04	full_name	String	NOT NULL	Full name of the user
05	is_active	Boolean	NOT NULL	Indicates if the user account is currently active
06	created_at	DateTime	NOT NULL	Timestamp when the account was created
07	Role	String	NOT NULL, Allowed Values: Admin, Student, Faculty	User role

2.2.2. Student

Sr. No	Name	Data Type	Constraint	Description
01	user_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier referencing the User entity
02	reg_number	String	NOT NULL, UNIQUE	University registration number of the student

2.2.3. Faculty

Sr. No	Name	Data Type	Constraint	Description
01	user_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier referencing the User entity
02	Designation	String	NOT NULL	Designation of Faculty (e.g. Lecturer, Professor)

2.2.4. Department

Sr. No	Name	Data Type	Constraint	Description
01	department_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the department
02	department_name	String	NOT NULL, UNIQUE	Name of the academic department

2.2.5. Program

Sr. No	Name	Data Type	Constraint	Description
01	program_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the program
02	program_name	String	NOT NULL, UNIQUE	Name of the academic program

2.2.6. Batch

Sr. No	Name	Data Type	Constraint	Description
01	batch_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the batch
02	batch_year	Integer	NOT NULL	The year the batch started

2.2.7. Semester

Sr. No	Name	Data Type	Constraint	Description
01	semester_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the semester
02	semester_name	String	NOT NULL	Name of the semester

2.2.8. Session

Sr. No	Name	Data Type	Constraint	Description
01	session_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the session
02	session_name	String	NOT NULL, UNIQUE, Allowed Values: Fall, Spring, Summer	Name of the academic session

2.2.9. Course

Sr. No	Name	Data Type	Constraint	Description
01	course_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the course
02	course_code	String	NOT NULL, UNIQUE	Official code of the course
03	course_name	String	NOT NULL	Title of the course
04	credit_hours	Integer	NOT NULL, Allowed Values: 1, 2, 3	Number of credit hours for the course

2.2.10. Venue

Sr. No	Name	Data Type	Constraint	Description
01	venue_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the venue

Sr. No	Name	Data Type	Constraint	Description
02	venue_name	String	NOT NULL, UNIQUE	Name or room number of the venue
03	venue_type	String	NOT NULL, Allowed Values: Lecture Hall, Auditorium, Lab	Type of venue
05	location	String	NULL	Building of the venue

2.2.11. Timeslot

Sr. No	Name	Data Type	Constraint	Description
01	timeslot_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the timeslot
02	day_of_week	String	NOT NULL, Allowed Values: Monday, Tuesday, Wednesday, Thursday, Friday	Day of the week
03	start_time	Time	NOT NULL	Start time of the slot
04	end_time	Time	NOT NULL	End time of the slot

2.2.12. Schedule Change Request

Sr. No	Name	Data Type	Constraint	Description
01	request_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the request
02	reason	String	NULL or Maximum 200 characters	Reason for the change request
03	status	String	Allowed Values: Pending, Approved, Rejected	Current status
04	created_at	DateTime	NOT NULL	Timestamp when the request was created

Sr. No	Name	Data Type	Constraint	Description
05	reviewed_at	DateTime	NOT NULL	Timestamp when the request was reviewed

2.2.13. Section

Sr. No	Name	Data Type	Constraint	Description
01	section_id (Primary Key)	Integer	NOT NULL, UNIQUE	Unique identifier for the section
02	section_name	String	NOT NULL	Name or label of the section

2.2.14. COURSE_OFFERING

Sr. No	Name	Data Type	Constraint	Description
01	Offering_id	Integer	NOT NULL, UNIQUE	Unique identifier for the course offering

2.2.15. Schedule

Sr. No	Name	Data Type	Constraint	Description
01	schedule_id	Integer	NOT NULL, UNIQUE	Unique identifier for the class schedule

2.3. BUSINESS RULES:

- A user must be assigned a specific role (Admin, Faculty, or Student) to access the system, enforcing Role-Based Access Control.
- A faculty member can be assigned to teach multiple course offerings but cannot be scheduled for two different course offerings during the exact same timeslot.
- A venue can host multiple scheduled classes throughout the week but cannot be double-booked for different course offerings at the exact same timeslot.
- A student can enroll in multiple course offerings, and a single course offering can accommodate multiple enrolled students.

- A schedule change request can only be submitted by the specific faculty member who is assigned to the corresponding course offering.
- All schedule change requests must be reviewed and approved or rejected.
- A course offering must be strictly associated with a valid course, faculty member, batch, session, and semester.
- A department can offer multiple courses, and a single course can be associated with multiple departments.
- A course offering may be divided into multiple sections, and each section must have an independent schedule.
- A schedule cannot exist without a valid course offering.
- A faculty member cannot be assigned overlapping schedules across different course offerings.
- A schedule change request can only be submitted by the faculty member assigned to the corresponding schedule.
- Each schedule change request must be linked to exactly one schedule entry and cannot modify multiple schedules simultaneously.
- A schedule change request must follow a defined status workflow.
- A user account must remain active for access to system functionalities.

2.4. ENTITY RELATIONSHIP DIAGRAM:

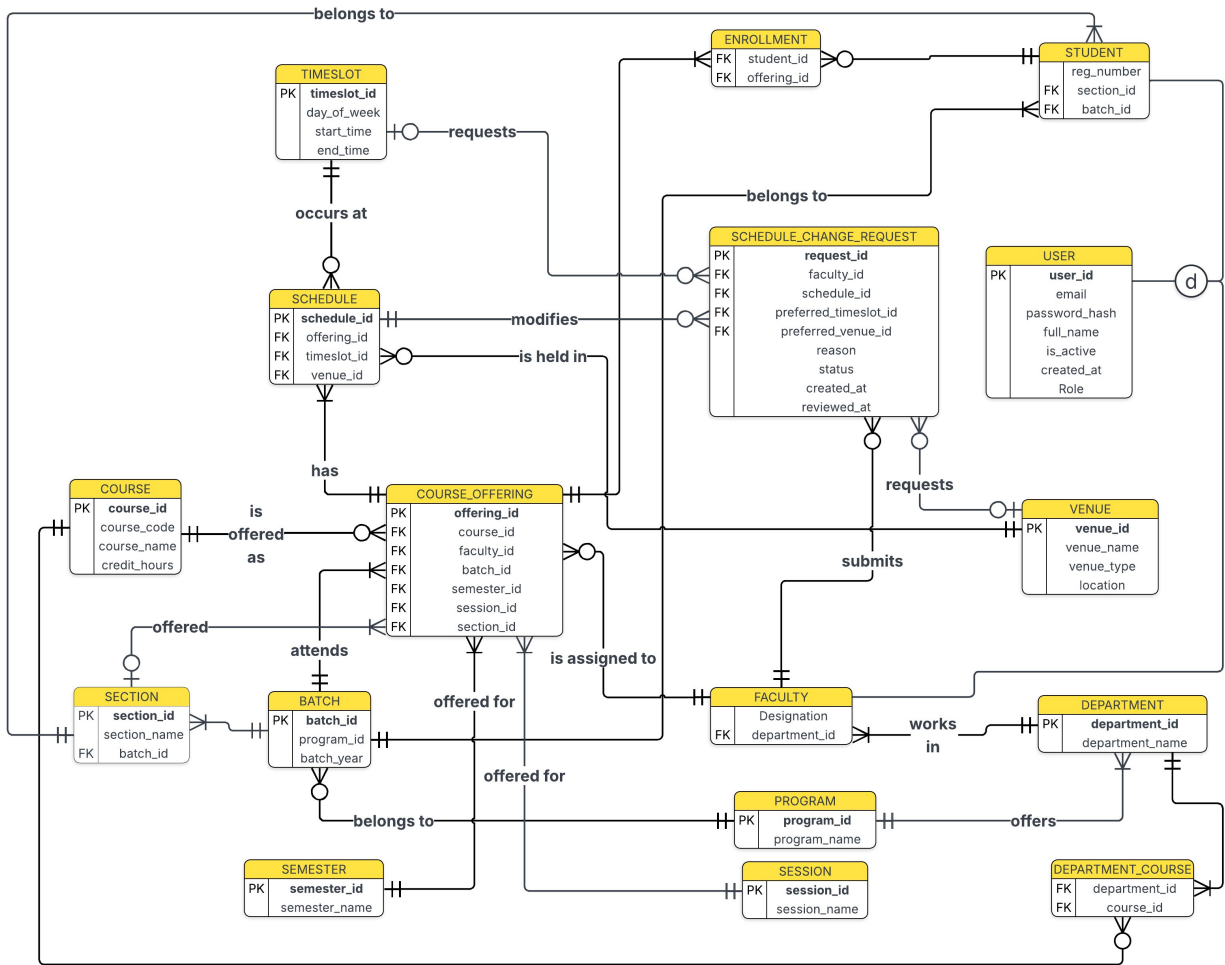


Figure 1: Entity-Relationship Diagram

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- [3] J. Nwankwo, “Crow's Foot Notation Relationship Symbols and How to Read Diagrams,” freeCodeCamp, Jul. 19, 2024. [Online]. Available: <https://www.freecodecamp.org/news/crows-foot-notation-relationship-symbols-and-how-to-read-diagrams/>.
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APPENDIX: AI USAGE AND PROMPTS

Sr. No	Purpose	Prompt
1	ERD Evaluation	“You are a database design expert and academic evaluator. I will provide you with my ERD (based on Crow’s Foot notation). Your task is to critically analyze, verify, and improve it according to database design best practices and academic grading standards...”
2	ERD Evaluation	“i have removed student-course entity, verify: 1. which student is taking which course 2. which department is offering which course 3. which faculty is teaching which course 4. i should be able to retrieve timetable of specific batch in specific session (spring 2025) of specific program 5. i should be able to retrieve timetable of specific semester of specific program specific batch in specific session”
3	ERD Enhancement	“course can be offered in multiple departments now, provide all changes to me including: 1. proper entity name 2. proper attributes 3. proper relationships with cardinality with participation 4. proper relationship names”
4	Lucidchart Prompt	“it is writing key name with attribute too, i dont want that. it is writing constraint as attributes, provide efficient prompt according to lucid chart ai, which can convert my text into erd”
5	Relationship Naming	“can you help me with naming the relationships”
6	Clarification	“is course offering associative entity?”
7	Relationship Naming	“name of relationship between student and program”
8	Example Data Generation	“example data according to my erd”
9	Workflow Verification	“can you verify the schedule change request workflow”
10	Design Validation	“Verify this in terms of all the things i said to you before like can i retrieve timetable for specific batch of specific program in specific session”